

Maths 141H – Homework 2

Due: **Beginning of class on 10/21/2024**

Rules

- This homework consists of **7 questions**. Please try to answer all the questions. You will be marked on the **best 6 out of 7 questions you answer**. If you attempt all seven questions, your grade on the question on which you have scored the least will be **extra credit** (again, it's possible to get more than 100% on this HW).
- Some questions have parts marked as (**Extra**). You don't have to answer these to get full marks, but correct answers to these parts will be rewarded.
- Work in groups of **2-3 people**. Each group should submit **a single solution**, clearly indicating the group members (names and UIDs are required). **Groups must have been decided by 10/10** and announced to me by email, before the end of the working day. If you cannot find a group, let me know. **All members of the same group will receive the same grade.**
- Submit your answers at the beginning of class on 10/21/2024. Ensure your answers are legible (illegible parts will be ignored), and to help with the **markinggrading**,¹ please answer each question on a fresh page and staple everything together.
- **Good luck!**

¹A happy **markergrader** is a generous marker! Keep this in mind.

Question 1 – Inverse Functions

1. Assume that f has an inverse, that $f''(f^{-1}(x))$ exists and that $f'(f^{-1}(x)) \neq 0$. Show that:

$$(f^{-1})''(x) = \frac{-f''(f^{-1}(x))}{[f'(f^{-1}(x))]^3}.$$

2. Suppose that the domain of a real-valued function f is an open interval I and that f is invertible on I . Suppose that both f' and f'' exist on I and that $f'(x) \neq 0$ and $f''(x) \neq 0$, for all $x \in I$.

- (a) Suppose that f is strictly increasing on I . Show that the graph of f^{-1} is concave up on its domain if the graph of f is concave down on I and that the graph of f^{-1} is concave down on its domain if the graph of f is concave upward on I .
- (b) Suppose that f is decreasing on I . Show that the graph of f^{-1} is concave up on its domain if the graph of f is concave up on I and that the graph of f^{-1} is concave down on its domain if the graph of f is concave down on I .
- (c) (**Extra**) Suppose that $a \in I$ and the graph of f has an inflexion point at $(a, f(a))$. What can you say about an inflexion point for the graph of f^{-1} ?
3. Let $0 \leq a \leq b$ and f a non-negative increasing continuous function on $[a, b]$. Let A_1 be the area of the region bounded by the graph of f and the x -axis, from $x = a$ to $x = b$, and A_2 the area of the region bounded by the graph of f and the y -axis, from $y = f(a)$ to $y = f(b)$.

- (a) Show that

$$A_1 + A_2 = bf(b) - af(a).$$

[*Hint*. Draw a picture!]

- (b) Using the previous part or otherwise prove that:

$$\int_a^b f(x)dx = bf(b) - af(a) - \int_{f(a)}^{f(b)} f^{-1}(y)dy.$$

- (c) Using the previous part, or otherwise, evaluate:

$$\int_0^1 \left((x-1)^{\frac{1}{3}} + 1 \right)^{\frac{1}{2}} dx.$$

Question 2 – More Inverse Functions

Recall that the hyperbolic cosine function is defined as follows:

$$\begin{aligned}\cosh : \mathbb{R} &\rightarrow \mathbb{R}_{>0} \\ x &\mapsto \frac{e^x + e^{-x}}{2}\end{aligned}$$

1. Show that $\cosh$ does not have an inverse.
2. Show that if we restrict the domain of $\cosh$ to $[0, \infty]$ then, the resulting function does have an inverse, $\cosh^{-1}(x)$.
3. Find the domain and range of $\cosh^{-1}(x)$.
4. Prove that:

$$\cosh^{-1} x = \ln \left(x + \sqrt{x^2 - 1} \right).$$

In so doing, explain where you use the fact that the restricted domain of $\cosh$ is $[0, \infty)$.

5. Use part (b) to prove that the following formulas are valid:

(a) $\sinh^{-1}(\sqrt{x^2 - 1}) = \cosh^{-1}(x)$ for $x \geq 1$.

(b) $\cosh^{-1}(\sqrt{x^2 + 1}) = \sinh^{-1}(x)$ for $x \geq 0$.

6. Let:

$$f(x) = c \cosh \left(\frac{x}{c} \right),$$

for $c \in \mathbb{R}_{>0}$, and $x \in [0, \infty)$. Show that:

$$f^{-1}(x) = c \ln \left(\frac{x + \sqrt{x^2 - c^2}}{c} \right),$$

for $x \geq c$.

7. Show that

$$\int \frac{1}{\sqrt{x^2 - 1}} dx = \cosh^{-1}(x) + C.$$

8. **(Extra)** Does $\tanh(x)$ have an inverse? If it does find the domain and the range of the inverse, otherwise explain why it does not.

Question 3 – L'Hôpital's Rule

1. Find:

$$\lim_{x \rightarrow 0} \frac{\int_0^x (x-t) \sin(t^2) dt}{\ln(1+x^4)}.$$

2. Let $f(x) = c \cosh\left(\frac{x}{c}\right)$, for $c \in \mathbb{R}_{>0}$. Find:

$$\lim_{x \rightarrow \infty} \frac{f^{-1}(x)}{c \ln(2x)}.$$

[*Hint.* It may be useful to look back at Question 2(6).]

3. Let f and g be defined on $(0, \infty)$. We say that f is of **higher order of magnitude** than g if

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \infty.$$

We denote this by $g \ll f$ (or, to the same extent, $f \gg g$). In addition, we say that f and g have **equal order of magnitude** if

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} \in \mathbb{R}_{>0}.$$

We denote this by $f \sim g$. Consider the following ten functions:

$$x^x, x^k, x^{\frac{1}{m}}, (\ln(x))^n, e^x, \cosh(x), \sinh(x), a^x, b^x, \text{fac}(x),$$

where $k, m, n \in \mathbb{N}_{>0}$, $a, b \in \mathbb{R}$ are such that $1 < a < e < b$, and

$$\text{fac}(x) := 1 \times 2 \times 3 \times \cdots \times \lceil x \rceil.$$

(Here $\lceil x \rceil$ denotes the least integer greater than or equal to x .)

Given this:

- Rank the first 9 functions from least to highest order of magnitude, indicating any that are of equal order or magnitude.
- (**Extra**) Where does **fac** place in the ranking from part (a)? (An educated guess will suffice.)
- Find a function f with a higher order of magnitude than all the functions from (a).
- Find a *positive* function g with a lower order of magnitude than all the functions from (a).

Question 4 – Inverse Trigonometric Functions

1. Prove that:

$$\sin^{-1}(x) + \sin^{-1}(y) = \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right).$$

2. Let $a, b > 0$. Evaluate:

$$\int \frac{1}{a^2 \sin^2(x) + b^2 \cos^2(x)} dx.$$

[*Hint.* Use the substitution $u = \tan(x)$.]

3. Prove that:

$$\sin^{-1}\left(\frac{x}{3} - 1\right) = \frac{\pi}{2} - 2\sin^{-1}\left(\sqrt{1 - \frac{x}{6}}\right)$$

and

$$\sin^{-1}\left(\frac{x}{3} - 1\right) = 2\sin^{-1}\left(\frac{\sqrt{x}}{\sqrt{6}}\right) - \frac{\pi}{2}.$$

4. Prove that if $x \in [-1, 1]$ there is some $c \in \mathbb{R}$ such that:

$$\sin^{-1}(x) + \cos^{-1}(x) = c.$$

[*Hint.* Consider differentiating the left-hand side.]

5. Determine the value of c , from the previous part.

6. (**Extra**) Let $f(x) = \tan^{-1}\left(\frac{x+1}{x-1}\right)$, for $x \neq 1$. Find $f'(x)$ and $f''(x)$. Prove that $\lim_{x \rightarrow 1^+} f(x) = \frac{\pi}{2}$, $\lim_{x \rightarrow 1^-} f(x) = -\frac{\pi}{2}$. With all this information, sketch the graph of f . How is $f(x)$ related to $g(x) = -\tan^{-1}(x)$?

Question 5 – More Inverse Trigonometric Functions

1. Let $x, y \in \mathbb{R}$ be such that $xy \neq 1$. Prove that:

$$\tan^{-1}(x) + \tan^{-1}(y) = \tan^{-1}\left(\frac{x+y}{1-xy}\right),$$

provided that the expression on the left-hand side is in $(-\frac{\pi}{2}, \frac{\pi}{2})$.

2. Using the previous part prove that:

(a) $\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \frac{\pi}{4}$.

(b) $2\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) = \frac{\pi}{4}$.

(c) $\tan^{-1}\left(\frac{120}{119}\right) - \tan^{-1}\left(\frac{1}{239}\right) = \frac{\pi}{4}$.

(d) $4\tan^{-1}\left(\frac{1}{5}\right) - \tan^{-1}\left(\frac{1}{239}\right) = \frac{\pi}{4}$

3. Suppose that $a, b, c \in \mathbb{R}$ are such that $a^2 + 1 = bc$. Show that:

$$\tan^{-1}\left(\frac{1}{a+b}\right) + \tan^{-1}\left(\frac{1}{a+c}\right) = \tan^{-1}\left(\frac{1}{a}\right),$$

provided that the expression on the left-hand side is in $(-\frac{\pi}{2}, \frac{\pi}{2})$ and none of the denominators above are 0.

4. Using the previous part show (again) that:

$$\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{3}\right) = \frac{\pi}{4}.$$

Question 6 – Integration By Parts

1. Prove the following:

(a) For $n \in \mathbb{N}_{\geq 2}$:

$$\int \cos^n(x) dx = \frac{1}{n} \cos^{n-1}(x) \sin(x) + \frac{n-1}{n} \int \cos^{n-2}(x) dx.$$

(b) For $n \in \mathbb{N}_{>0}$:

$$\int (\ln(x))^n = x(\ln(x))^n - n \int (\ln(x))^{n-1} dx.$$

(c) For all $n \in \mathbb{N}$:

$$\int \frac{1}{(x^2 + a^2)^{n+1}} dx = \frac{x}{2na^2(x^2 + a^2)^n} + \frac{2n-1}{2na^2} \int \frac{1}{(x^2 + a^2)^n} dx.$$

(d) (**Extra**) For $n \in \mathbb{N}_{\geq 2}$:

$$\int \sec^n(x) dx = \frac{\sec^{n-2}(x) \tan(x)}{n-1} + \frac{n-2}{n-1} \int \sec^{n-2}(x) dx.$$

2. Using the previous parts, or otherwise, evaluate the following integrals:

(a)

$$\int (\ln(x))^3 dx.$$

(b)

$$\int_{\frac{\pi}{3}}^{\frac{\pi}{4}} \sec^5(x) dx.$$

3. Let f be a differentiable function on some interval $[a, b]$.

(a) Assuming that f has a continuous second derivative on $[a, b]$ show that:

$$f(b) = f(a) + f'(a)(b-a) + \int_a^b f''(x)(b-x) dx.$$

(b) Assuming that f has a continuous third derivative on $[a, b]$ show that:

$$f(b) = f(a) + f'(a)(b-a) + \frac{f''(a)}{2}(b-a)^2 + \frac{1}{2} \int_a^b f^{(3)}(x)(b-x)^2 dx.$$

Question 7 – Trigonometric Integrals

1. Let $f(x) = \sqrt{1-x^2}$ and $g(x) = -(1+x)$. Find the centre of gravity of the region R between the two graphs, on $[0, 1]$.
2. Let $f(x) = x^2 - x$, for $x \in [0, 1]$. Find the length of the graph of f .
3. Let $f(x) = x^2$, for $x \in [0, \frac{1}{2}]$. Find the length of the graph of f .
4. Evaluate:

$$\int \left(\frac{1-x}{1+x} \right)^{\frac{1}{2}} dx.$$

[*Hint* Consider using the substitution $x = \cos(2u)$.]

5. Let $m, n \in \mathbb{N}_{>0}$.

(a) Prove that:

$$\int_{-\pi}^{\pi} \sin(mx) \cos(nx) dx = 0.$$

(b) Prove that:

$$\int_{-\pi}^{\pi} \cos(mx) \cos(nx) dx = \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{otherwise.} \end{cases}.$$

(c) Prove that:

$$\int_{-\pi}^{\pi} \sin(mx) \sin(nx) dx = \begin{cases} 0 & \text{if } m \neq n \\ \pi & \text{otherwise.} \end{cases}.$$